SPARKY'S SCIENCE FACTS

Why Does the Moon Change Shape?

Astronomy · Ages 7+

> **Short answer:** The Moon does not really change shape. We just see different amounts of its sunny side as it travels around the Earth.

How it works

- The Moon makes no light of its own. It shines because sunlight lights it up.
- Half of the Moon is always lit by the Sun, like half of a ball in torchlight.
- The Moon goes around the Earth about once a month: from one new moon to the next takes about 29 and a half days.
- At new moon we see none of the sunny side, at full moon all of it, and crescent and half moons in between. These are the phases of the Moon.

Key words: Phases: the shapes the Moon seems to have; Waxing and waning: growing and shrinking

Common mix-ups

- The phases are not caused by the Earth's shadow. The Earth's shadow on the Moon is a lunar eclipse, which happens only now and then, at full moon.
- In South Africa the growing Moon is lit on the left. Pictures from the north show it on the right.

Did you know? Many farmers in Southern Africa have long used the Moon's phases to count the months.

Try it: In a dark room, hold a ball at arm's length near a lamp and turn slowly around. Watch the lit part change, just like the Moon!

Ask Sparky: Why can we sometimes see the Moon in the day? What is a lunar eclipse?